

HELIOS OBSERVABILITY PLATFORM

Service: api-gateway**Severity:** high

Incident Report: anomaly_api-gateway_1779360609

Service: api-gateway**Detected:** 2026-05-21T10:50:09.156168**Severity:** HIGH**Model confidence:** 0.95

Executive summary

The api-gateway service is experiencing a high-severity incident characterized by a significant spike in error counts and extreme latency degradation. Sample logs indicate the service is failing due to recurring database timeouts.

Root cause hypothesis

The api-gateway is suffering from a cascading failure driven by database connectivity issues. The SHAP attribution shows that `error_count` and `p99_latency_ms` are the primary drivers of the anomaly, consistent with the observed 'DB timeout' logs. The high p95 and p99 latency values suggest that queries are queuing or timing out at the database layer, leading to a high error rate of 0.489.

Top contributing features (SHAP)

Feature	Value	SHAP	Direction
error_count	489.000	-1.628	toward_anomaly
p99_latency_ms	6395.605	-1.621	toward_anomaly
p95_latency_ms	6101.888	-1.397	toward_anomaly

Recommended actions

Immediate

- **Verify database health and connectivity status for the api-gateway service.** — Sample logs explicitly report 'DB timeout' errors, suggesting the root cause is a backend database bottleneck.

Short Term

- **Review recent database schema changes or query performance regressions.** — The extreme p99_latency_ms values (6395ms) suggest that long-running queries are likely causing the observed timeout errors.

Long Term

- **Implement circuit breakers for database-dependent operations.** — The high error_count indicates that the service is currently overwhelmed by failing requests, which could be mitigated by failing fast during database outages.

Monitoring follow-up

- api_gateway_error_count
- api_gateway_p99_latency_ms
- database_connection_pool_saturation
- database_query_execution_time

Generated by Helios at 2026-05-21T10:50:31+00:00

This is an automated incident report generated by Helios Observability Platform
For questions or concerns, contact your DevOps team